

## Finite Product Spaces

In this section, we focus on product spaces of the form  $X \times Y$ , where  $X$  and  $Y$  are topological spaces. With that foundation secured, products of the form

$$X_1 \times X_2 \times \cdots \times X_n$$

will be easy to understand.

### Definition

### Product Space

Let  $(X, T)$  and  $(Y, U)$  be topological spaces. The **product space** of  $X$  and  $Y$  as

$$X \times Y = \{(x, y) : x \in X, y \in Y\}$$

with basic open sets of the form

$$U \times V \text{ where } U \text{ is open}_X \text{ and } V \text{ is open}_Y$$

Indeed, these basic open sets generate a topology. First,  $X \times Y$  is itself basic open since  $X$  is  $\text{open}_X$  and  $Y$  is  $\text{open}_Y$ . Moreover, if  $U_1, U_2$  are  $\text{open}_X$  and  $V_1, V_2$  are  $\text{open}_Y$ ,

$$(U_1 \times V_1) \cap (U_2 \times V_2) = (U_1 \cap U_2) \times (V_1 \cap V_2)$$

so the intersection of two basic open sets is itself an open set. Hence this base generates the product topology on  $X \times Y$ .

Open sets on the product topology can also be defined point-wise. A set  $W \subseteq X \times Y$  is open if and only if for every point  $(x, y) \in W$ , there exist  $U \text{ open}_X$  and  $V \text{ open}_Y$  such that

$$(x, y) \in U \times V \subseteq W.$$

### Definition

### Projection Map

Let  $X \times Y$  be the product space of topological spaces  $(X, T)$  and  $(Y, U)$ . The **projection maps**  $\pi_X$  and  $\pi_Y$  are defined as

$$\pi_X : X \times Y \rightarrow X \quad (x, y) \mapsto x$$

$$\pi_Y : X \times Y \rightarrow Y \quad (x, y) \mapsto y$$

### Theorem

Suppose  $X \times Y$  is a product space. Then the topology on  $X \times Y$  is the “smallest”—that is, having the fewest open sets—so that the projection maps are continuous.

*Proof:* Suppose  $U \subseteq X$  is  $\text{open}_X$ . Then

$$\begin{aligned} (x, y) \in \pi_X^{-1}(U) &\iff \pi_X(x, y) \in U \\ &\iff x \in U \\ &\iff (x, y) \in U \times Y. \end{aligned}$$

So  $\pi_X^{-1}(U) = U \times Y$ , which is basic open. Similarly, if  $V$  is  $\text{open}_Y$ , then  $\pi_Y^{-1}(V) = X \times V$ , which is also basic open. Hence  $\pi_X$  and  $\pi_Y$  are continuous. ■

### Theorem

Projection maps are open; that is, if  $W \subseteq X \times Y$  is open in  $X \times Y$ , then  $\pi_X(W)$  is open in  $X$ . Similarly,  $\pi_Y$  is open.

*Proof:* Suppose  $W \subseteq X \times Y$  is open in  $X \times Y$ . We prove that  $\pi_X(W)$  is open one point at a time.

Let  $x \in \pi_X(W)$ . Then  $x = \pi_X(x, y)$  where  $(x, y) \in W$  for some  $y \in Y$ . Because  $W$  is open,  $(x, y) \in U \times V \subseteq W$  for some  $\text{open}_X U$  and  $\text{open}_Y V$ .

Hence  $x \in U \subseteq \pi_X(W)$ , meaning  $\pi_X(W)$  is  $\text{open}_X$ . Similarly,  $\pi_Y(W)$  is  $\text{open}_Y$ . ■

### Theorem

Suppose  $X$ ,  $Y$ , and  $Z$  are topological spaces and let  $f : X \times Y \rightarrow Z$  be a function. Then  $f$  is continuous if and only if  $\pi_X \circ f : Z \rightarrow X$  and  $\pi_Y \circ f : Z \rightarrow Y$  are continuous.

*Proof:* If  $f$  is continuous, then the result follows from the continuity of composing continuous maps.

Conversely  $\pi_X \circ f$  and  $\pi_Y \circ f$  are continuous. Let  $U \times V$  be basic open in  $X \times Y$ . So

$$\begin{aligned} z \in f^{-1}(U \times V) &\iff f(z) \in U \times V \\ &\iff (\pi_X \circ f)(z) \in U \text{ and } (\pi_Y \circ f)(z) \in V \\ &\iff z \in (\pi_X \circ f)^{-1}(U) \text{ and } z \in (\pi_Y \circ f)^{-1}(V). \end{aligned}$$

Hence  $f^{-1}(U \times V) = (\pi_X \circ f)^{-1}(U) \cap (\pi_Y \circ f)^{-1}(V)$ , which is an intersection of open sets and thus open.

We've shown  $f$  is “basically continuous,” as  $f^{-1}(B)$  for basic open  $B$  is open. Any open set is a union of basic open sets  $B_a$ , so

$$f^{-1}\left(\bigcup B_a\right) = \bigcup f^{-1}(B_a)$$

which is a union of open sets. So  $f$  being “basically continuous” implies it is indeed continuous. ■

## Quotient Spaces

Let  $X$  be a topological space, and let  $\sim$  be an equivalence relation on  $X$ . One idea is to “glue” equivalent parts of  $X$  together.

Let  $Y = X/\sim$ . A point in  $Y$  corresponds to an equivalence class in  $X$ . We define

$$\pi : X \rightarrow Y \quad x \mapsto [x]_{\sim}$$

to be a function mapping  $x \in X$  to its equivalence class in  $Y$ .

If we define an  $\text{open}_Y$  set  $V$  such that  $\pi^{-1}(V)$  is  $\text{open}_X$ , we endow  $Y$  with the **quotient topology**.

### Example

Taking a square and “gluing” its parallel edges is homeomorphic to a torus.